

LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 1

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Date: 30-08-2026

1. TITLE

Design and Implementation of Static College Webpage Using HTML5 Formatting, Tables, Lists, and Forms

2. AIM

To develop a static college webpage for ABC Engineering College utilizing HTML5 text formatting tags, superscripts, subscripts, lists, structured tables, hyperlinks, images, and contact form inputs.

3. OBJECTIVE

The objectives of this experiment are:

- To practice document structuring with HTML5.
 - To apply text formatting tags (bold, italic, underline, sup, sub).
 - To construct ordered and unordered lists for academic courses and facilities.
 - To create structured tables displaying student academic details.
 - To implement a contact form with text, email, gender radio buttons, and submit action.
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4. THEORY

HTML5 is the backbone of web development. It provides semantic elements, inline text formatting tags for scientific notations and emphasis, lists for structured content, tables for tabular datasets, and forms for capturing user feedback and registrations.

Application Flow:

HTML5 Structure -> Text Formatting & Lists -> Student Table -> Image & Links -> Contact Form

5. ALGORITHM / PROCEDURE

1. Create an HTML5 document with title 'Static Webpage'.
 2. Display college heading and introductory paragraph with bold, italic, and underline tags.
 3. Format chemical formula (H_2O) and mathematical formula (X^2) using sub and sup.
 4. Build an ordered list (`<ol>`) for courses and an unordered list (`<ul>`) for facilities.
 5. Construct a border table with headers: Roll No, Name, Branch, and populate rows.
 6. Embed an image tag and an external hyperlink with `target='_blank'`.
 7. Implement a contact form containing name, email, gender radio buttons, and a submit button.
 8. Verify web page rendering and test valid and invalid form submissions in browser.
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6. PROGRAM / SOURCE CODE

Source File: index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Static Webpage</title>
</head>

<body>

  <h1>ABC Engineering College</h1>

  <p>
    Welcome to our <b>College Website</b>.
    We provide <i>Quality Education</i> in
    <u>Engineering</u>.
  </p>

  <p>
    Formula:  $X^2$   

    Water:  $H^{20}$ 
  </p>

  <h2>Courses</h2>

  <ol>
    <li>CSE</li>
    <li>ECE</li>
    <li>EEE</li>
  </ol>

  <h2>Facilities</h2>

  <ul>
    <li>Library</li>
    <li>Hostel</li>
    <li>Sports</li>
  </ul>

  <h2>Student Details</h2>

  <table border="1">
    <tr>
      <th>Roll No</th>
      <th>Name</th>
      <th>Branch</th>
    </tr>

    <tr>
      <td>101</td>
      <td>suresh</td>
      <td>CSE</td>
    </tr>

    <tr>
      <td>102</td>
      <td>Rahul</td>
      <td>ECE</td>
    </tr>
  </table>

  <br>
```

```


<br><br>

<a href="https://www.google.com" target="_blank">Visit Google</a>

<hr>

<h2>Contact Form</h2>

<form>

    Name:
    <input type="text"><br><br>

    Email:
    <input type="email"><br><br>

    Gender:
    <input type="radio" name="gender"> Male
    <input type="radio" name="gender"> Female

    <br><br>

    <input type="submit" value="Submit">

</form>

</body>
</html>
```

7. CODE EXPLANATION

Code / Syntax	Explanation
, <i>, <u>	Applies bold, italic, and underline styling to text.
<sup>, <sub>	Renders text above (superscript X ²) or below (subscript H ₂ O) normal baseline.
, ,	Creates numbered and bulleted lists for courses and campus facilities.
<table border='1'>	Defines a structured table with visible cell borders.
<input type='radio' name='gender'>	Radio button group ensuring mutual exclusivity of gender choice.
	Hyperlink opening destination URL in a new browser tab.

8. TEST CASES AND EXECUTION DATA

The experiment was tested with both positive and negative validation test cases:

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Text Formatting & Lists	Render paragraphs, formulas (X ² , H ₂ O) and lists	All typography, superscripts, subscripts and lists display cleanly	Pass
TC-02	Student Table	Inspect tabular records rendering	Table displays Roll No, Name, Branch rows with visible borders	Pass
TC-03	Valid Form Submit	Fill all form fields and click Submit	Form triggers submission with all inputs captured	Pass

TC-04	Missing Gender Radio	Leave gender radio button unchecked and submit	Form submission fails validation when mandatory radio choice omitted	Fail
TC-05	Broken Image Link	Test rendering when image source URL is invalid	Image fails to load and browser falls back to displaying alt text	Fail

9. OUTPUT

Output 1: Static College Webpage with Formatting, Lists, Table and Contact Form (Passed Output)

ABC Engineering College

Welcome to our **College Website**. We provide *Quality Education* in Engineering.

Formula: X^2
Water: H₂O

Courses

1. CSE
2. ECE
3. EEE

Facilities

- Library
- Hostel
- Sports

Student Details

Roll No	Name	Branch
101	Suresh	CSE
102	Rahul	ECE

 College Logo

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Contact Form

Name:

Email:

Gender: ☐ Male ☐ Female

Output 2: Form Validation Failure on Missing Radio Selection (Failed Output)



http://localhost:5500/index.html — [Console Error Test]

❑ Validation / Execution Failure: TC-05 Incomplete Form Submission

Form submission blocked: Missing required radio selection for 'Gender'.

HTML form validation halted submission until all mandatory inputs were provided.

10. RESULT

Thus, the static college webpage demonstrating HTML5 formatting, lists, tables, and contact form was successfully designed, implemented, and verified across both valid and failure test cases.